BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Boolean Logic: AND, OR and NOT

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why should a corrected answer be retested with a different example?

2. What does an ethical issue concern?
   A) What is fair and right
   B) Only what is legal
   C) Only what is popular
   D) Only what is fastest

3. "Open source means there are never any licence conditions or costs." Correct this. Explain why.

4. A computer needs maintenance and data protection. "Utilities are only apps for writing essays." Correct this. What do utilities do?

Today's objectives

Explain the core ideas of boolean logic: and, or and not using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

AND gate
Output 1 only when both inputs are 1.

OR gate
Output 1 when at least one input is 1.

NOT gate
Reverses its single binary input.

Boolean expressions
Symbols and truth tables describe logic gate relationships independently of hardware.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

AND gate
Output 1 only when both inputs are 1.

OR gate
Output 1 when at least one input is 1.

NOT gate
Reverses its single binary input.

Boolean expressions
Symbols and truth tables describe logic gate relationships independently of hardware.

I Do - Worked model

For Q = A AND B, complete all four rows of the two-input truth table.

- List A,B as 00,01,10,11.
- AND returns 1 only if both are 1.
- Outputs are 0,0,0,1.
- Draw a two-input AND symbol matching the rule.

We Do - Guided practice

1. For Q = A OR B, complete the truth table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the Boolean test that denies access when authorisation is absent.

```
authorised = false
if ... then
print("Deny")
endif
```

You Do - Independent exam questions

Q2. State AND output when A=1, B=0. [1 mark]

Q3. Write OR outputs for inputs 00,01,10,11. [4 marks]

Q4. Explain NOT output for input 0. [2 marks]

Exit Ticket - True or False?

An AND gate outputs 1 only when all its inputs are 1.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. To check that the underlying method works independently on another case.

2. A - What is fair and right

3. It permits access to source code and modification or redistribution under its conditions.

4. Utilities maintain or protect a computer, adding tools beyond the OS. Writing an essay is an application task.

I Do - Explain the model

List A,B as 00,01,10,11.

AND returns 1 only if both are 1.

Outputs are 0,0,0,1.

Draw a two-input AND symbol matching the rule.

We Do - Model answers

1. Outputs 0,1,1,1.

Code First - Answer

```
authorised = false
if NOT authorised then
print("Deny")
endif
```

Expected output: Deny

You Do - Self-mark

Q2: 0.

Q3: 0,1,1,1.

Q4: NOT inverts the bit; output 1.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.